A LAB PROJECT REPORT

CSE4039

L39+L40

ADVANCED TEXT AND SOCIAL MEDIA ANALYTICS

Submitted By:

Team : ASTRIX

Submitted To:

Gundimeda Venugopal

SCOPE



TEXT SUMMARIZATION AND CLUSTERING FOR CRICKET T20 WORLD CUP

Experiment: Text summarization and clustering for cricket T20 World Cup news.

Aim: To make a short summary of cricket news and group similar news items using Python.

Introduction: During the T20 World Cup, many news items are published about matches, players, teams and weather. It can take time to read all the news. In this lab, I used a small set of cricket news sentences and tried to summarize them and group similar items.

Procedure

1. Take a few sample news sentences related to the T20 World Cup.
2. Convert the text to lowercase and remove punctuation and common words.
3. Count important words and use them to select three sentences for the summary.
4. Convert the news sentences into numerical values using TF-IDF.
5. Use K-Means to group the news items into three clusters.
6. Print the summary and cluster number for each news item.

Program

```
import re
from collections import Counter
from sklearn.feature_extraction.text import TfidfVectorizer, ENGLISH_STOP_WORDS from sklearn.cluster import
KMeans

news = [
"India defeated Australia in a T20 World Cup match with a strong batting performance.", "The Indian bowlers
restricted Australia and helped India win the World Cup match.", "England advanced after a close T20 World Cup
game with a last over finish.",
"The captain praised the bowlers after a tense cricket match.", "Fans filled the stadium to watch the T20 World
Cup semi-final.",
"The opening batter scored quickly and helped the team post a big total.", "Rain affected the World Cup match and
officials checked conditions.", "The coach said the squad is preparing for the next T20 World Cup match."
]

def clean(text):
text = re.sub(r"[^a-zA-Z ]", " ", text.lower()) words = [w for w in text.split()
if w not in ENGLISH_STOP_WORDS and len(w) > 2] return " ".join(words)

# Make a simple summary all_text = " ".join(news)
sentences = re.split(r"(?<=[.!?])\\s+", all_text) freq = Counter(clean(all_text).split())

scores = []
for sentence in sentences:
score = sum(freq[w] for w in clean(sentence).split()) scores.append((score, sentence))

summary = [s for score, s in sorted(scores, reverse=True)[:3]] print("SUMMARY:")
for sentence in summary: print("-", sentence)

# Group similar news items
vectorizer = TfidfVectorizer(stop_words="english") matrix = vectorizer.fit_transform(news)
model = KMeans(n_clusters=3, random_state=42, n_init=10) labels = model.fit_predict(matrix)

print("\\nCLUSTERS:")
for i, item in enumerate(news):
print(f"News {i+1} -> Cluster {labels[i]+1}: {item}")
```

Working of the Program

1. **Importing libraries:** The program uses Python libraries for cleaning text, counting words and clustering news.

2. **News list:** The list contains eight short news items about match results, bowling, batting, fans, rain and team preparation.

3. **Cleaning:** The clean()function changes the text to lowercase, removes punctuation and removes common English words such as “the” and “and”.

4. **Summarization:** The program counts meaningful words and gives each sentence a score. The three sentences with the highest scores are printed as the summary. It selects sentences from the original text, so it is called extractive summarization.

5. **TF-IDF:** TF-IDF changes text into numerical values. Words that help distinguish one news item from another are given importance. These values are used by the clustering algorithm.

6. **K-Means:** K-Means separates the news items into three groups. News with similar words may be placed in the same cluster. The cluster numbers do not have a fixed meaning.

7. **Output:** The program first prints the summary and then prints each news item with its cluster number.

Sample Output

The output below shows the format. The exact cluster numbers may be different when the program is run.

```
SUMMARY:

-  India defeated Australia in a T20 World Cup match with a strong batting performance.
-  The Indian bowlers restricted Australia and helped India win the World Cup match.
-  England advanced after a close T20 World Cup game with a last over finish.

CLUSTERS:

News 1 -> Cluster 2: India defeated Australia in a T20 World
Cup match... News 2 -> Cluster 2: The Indian bowlers
restricted Australia...

News 3 -> Cluster 1: England advanced after a close T20 World
Cup game... News 4 -> Cluster 1: The captain praised the
bowlers...

News 5 -> Cluster 3: Fans filled the stadium to watch the T20
World Cup... News 6 -> Cluster 2: The opening batter scored
quickly...

News 7 -> Cluster 3: Rain affected the World Cup
match... News 8 -> Cluster 1: The coach said the
squad is preparing...
```

Result and Discussion

Result: The program demonstrates text summarization and clustering on sample cricket news. It prints three selected sentences as a summary and assigns every news item to one of three clusters.

Discussion: News about batting and match results may share words such as “match”, “India” and “World Cup”. Other items may focus on fans, weather or team preparation. TF-IDF helps represent these word patterns, and K-Means uses the patterns to make groups. The grouping is based on text similarity, so it may not always match the categories a person would choose.

Limitations:

1. Only eight sample news items are used, so the result is not suitable for making conclusions about the whole tournament.
2. The sample news is written in the program instead of being collected automatically from news websites.
3. Word-frequency summarization can select sentences that repeat common ideas and may miss important match details.
4. K-Means requires the number of clusters to be chosen in advance.

Possible Improvements

The project can be improved by collecting more news articles, removing duplicate articles, and separating match reports from player interviews. A larger dataset may produce more useful clusters. The summary can also be checked manually to see whether it includes the important information from the original reports. In a future version, the program could display the most common words in each cluster to make the groups easier to understand.

Applications

1. Quickly reading the main points from several cricket news items.
2. Grouping news related to match results, player performance, team updates and weather.
3. Making a simple cricket news dashboard.
4. Helping users find articles about similar topics.

Conclusion

In this lab, I used basic NLP methods to summarize and cluster cricket T20 World Cup news. Word frequency was used to select summary sentences, and TF-IDF with K-Means was used to group similar items. This experiment helped me understand how text can be cleaned, converted into numerical form and organized automatically. The current program is a simple demonstration and can be extended with a larger collection of real news.

To run: Install scikit-learn with `pip install scikit-learn` and run the Python file.